Project Scenario 1: Small Business Network

Requirements

The company has:

- Management Department (VLAN 10)
- Sales Department (VLAN 20)
- IT Department (VLAN 30)

Each department should:

- Be separated using VLANs
- Communicate with other VLANs through Inter-VLAN Routing
- Receive IP addresses automatically via DHCP
- Use a DNS server for name resolution

1. Objective

To design and implement a small business network using VLANs, trunking, inter-VLAN routing, DHCP and DNS services within Cisco Packet Tracer.

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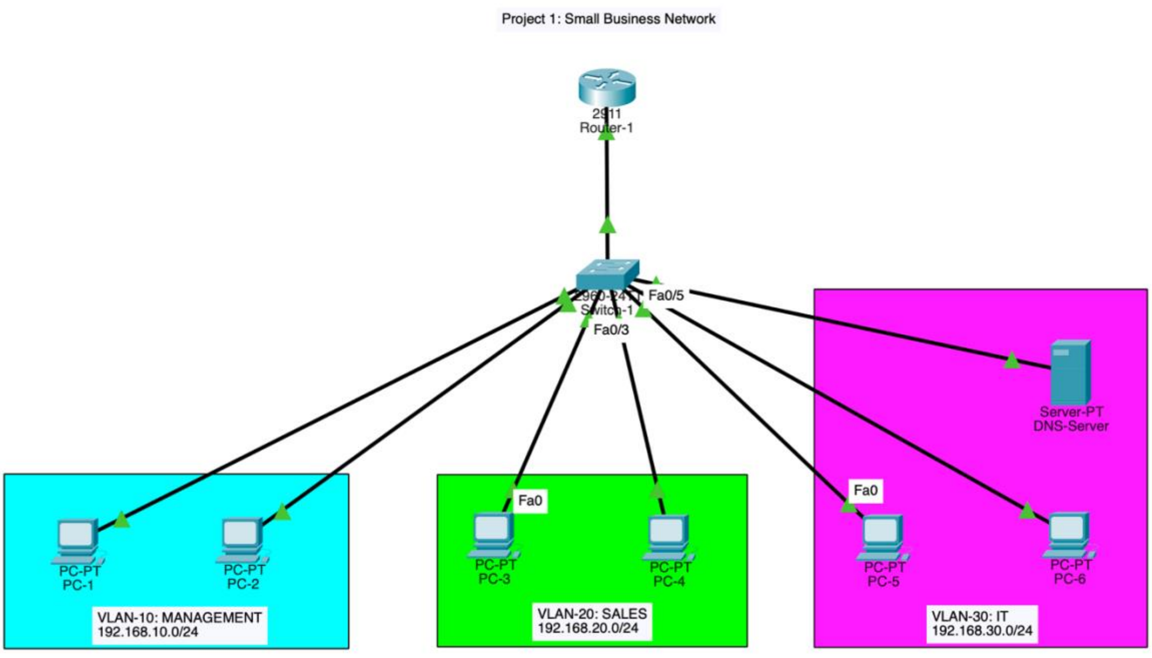
2. Network Topology



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3. IP Addressing Scheme

VLAN	Department	Network	Gateway
10	Management	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1
30	IT	192.168.30.0/24	192.168.30.1

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4. Switch Configuration

Create VLANS on Switch-1:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#
```

VLAN Brief:

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   MANAGEMENT                       active
20   SALES                            active
30   IT                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Assigning Ports to each VLAN:

Device	VLAN	Port
PC-1	10	Fa0/1
PC-2	10	Fa0/2
PC-3	20	Fa0/3
PC-4	20	Fa0/4

Device	VLAN	Port
PC-5	30	Fa0/5
PC-6	30	Fa0/6
DNS-Server	30	Fa0/7

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)# interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)# interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)# interface range fa0/5-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
```

Configure Trunking: The switch-to-router connection must carry all VLANs.

Since the router connected to interface g0/1:

```
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
```

Running-Config Verification:

```
Switch#show running-config
Building configuration...

Current configuration : 1409 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 30
 switchport mode access
!

interface GigabitEthernet0/1
 switchport mode trunk
!
interface GigabitEthernet0/2
!
```

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5. Router Configuration

Configure Inter-VLAN Routing: This is Router-on-a-Stick.

Router interface:

```
Router>en
Router# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown
```

Create sub interfaces:

```
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface g0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#
```

Verify with interface brief:

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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6. DHCP Configuration

The Router will provide DHCP. Exclude gateway addresses:

```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1
Router(config)#
```

DHCP VLAN Pools:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool MANAGEMENT
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#
```

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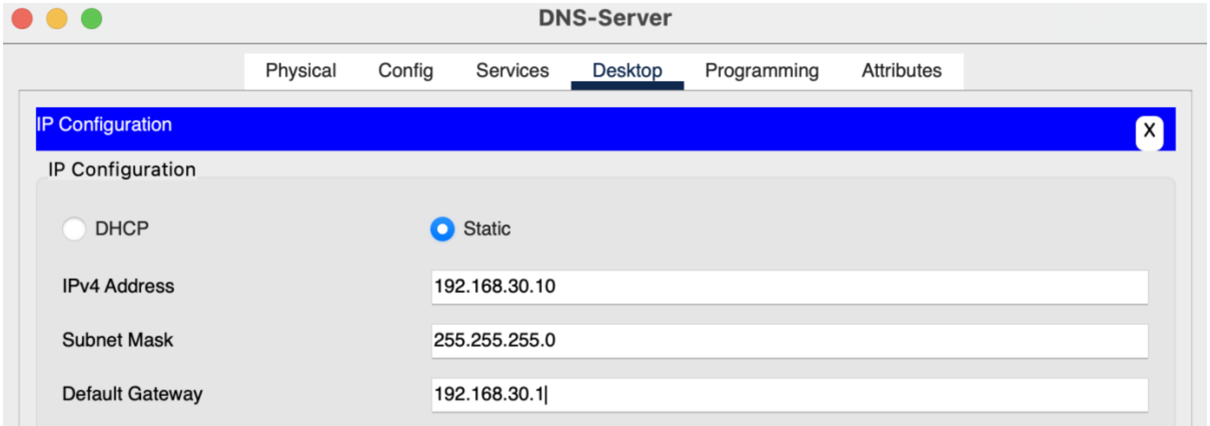
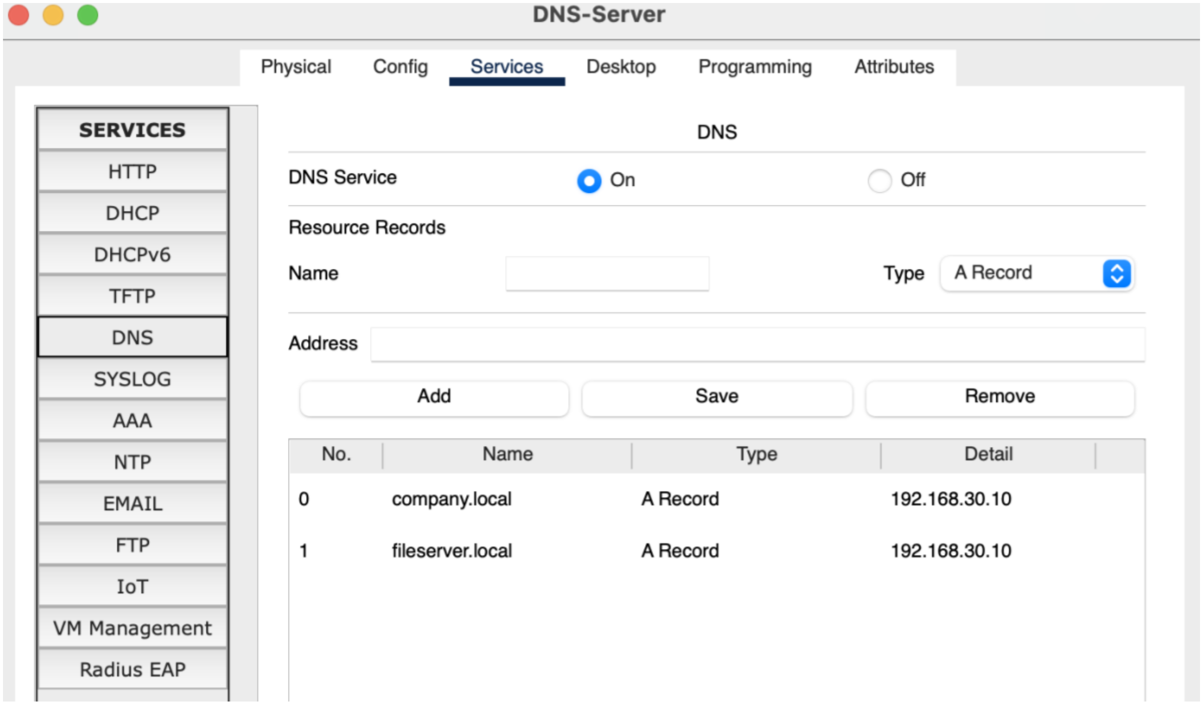
7. DNS Configuration

Desktop IP Configuration:

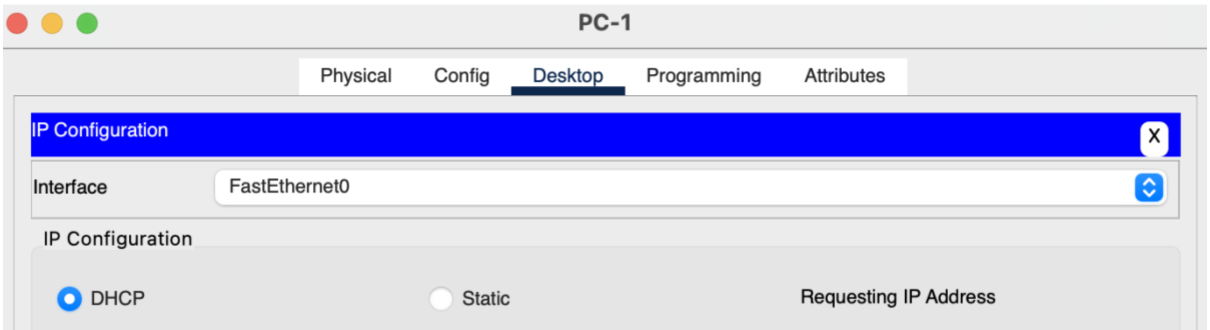


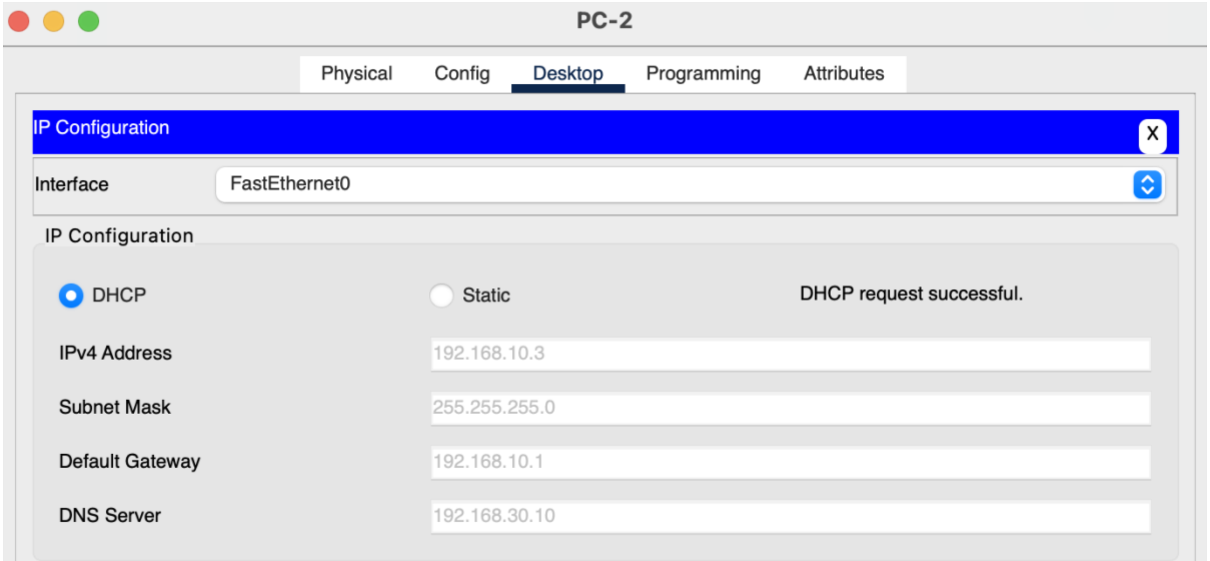
Service Records Config:



8. PC Configurations

All PC's are using DHCP for IP Configurations





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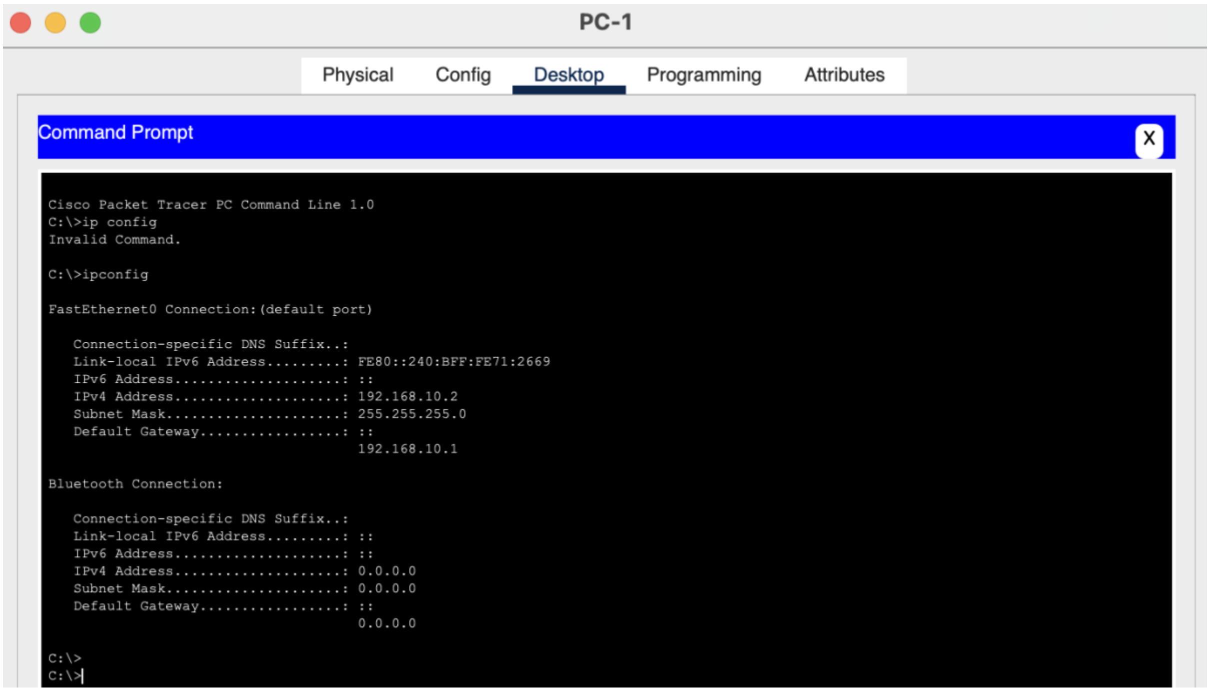
9. Testing Results

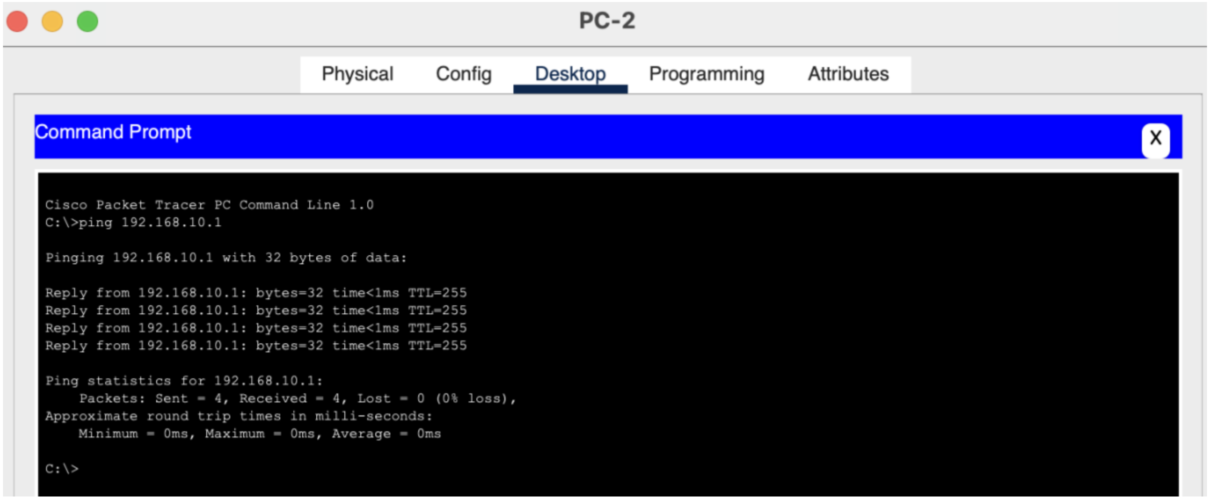
Include screenshots of:

Test 1 – DHCP:



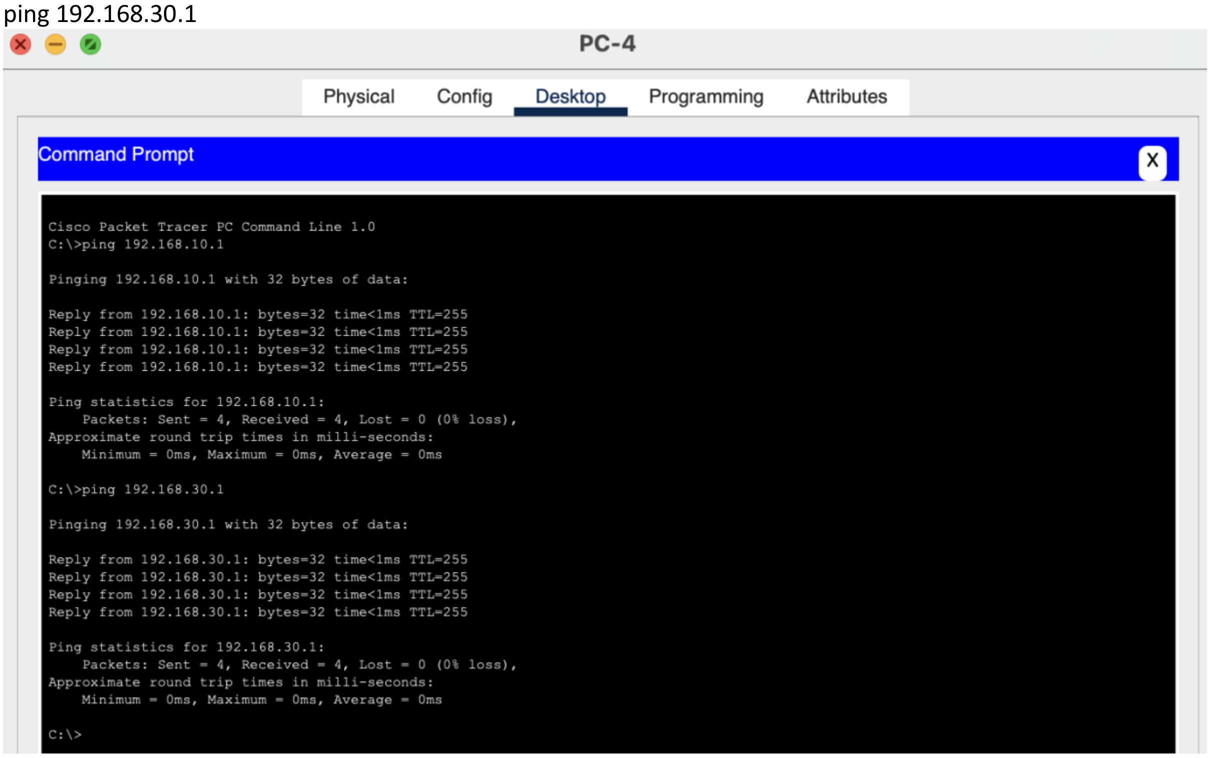
Result: Confirms correct subnet and IP Address.

Test 2 – Gateway:
ping 192.168.10.1



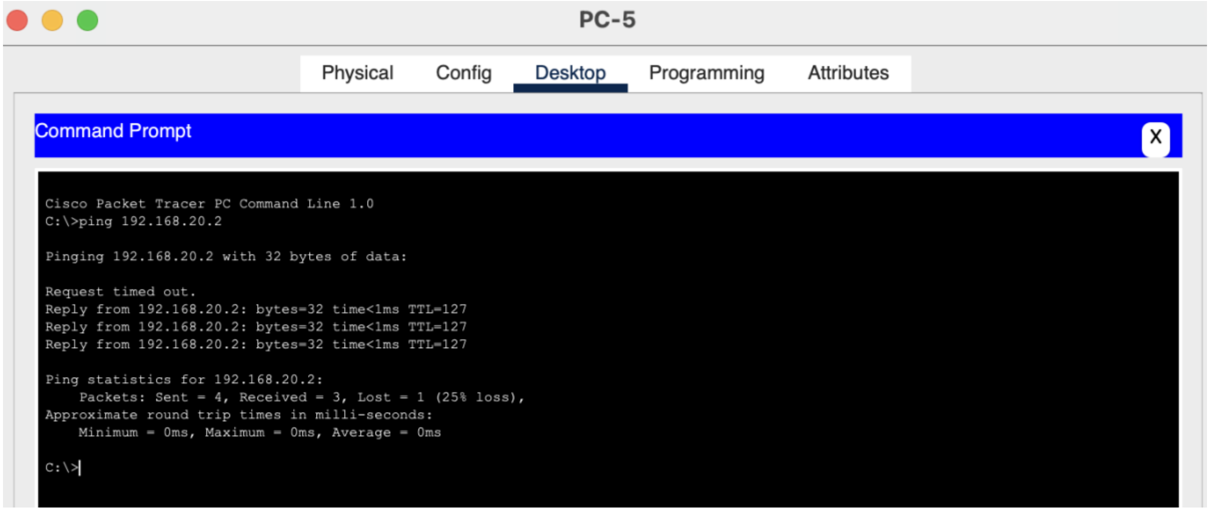
Result: All packets sent to default gateway with no issues.

Test 3 – Inter-VLAN Routing:
From VLAN 20 (SALES) PC:
ping 192.168.10.1
ping 192.168.30.1



Result: PCs in each VLAN are able to ping the other default gateways

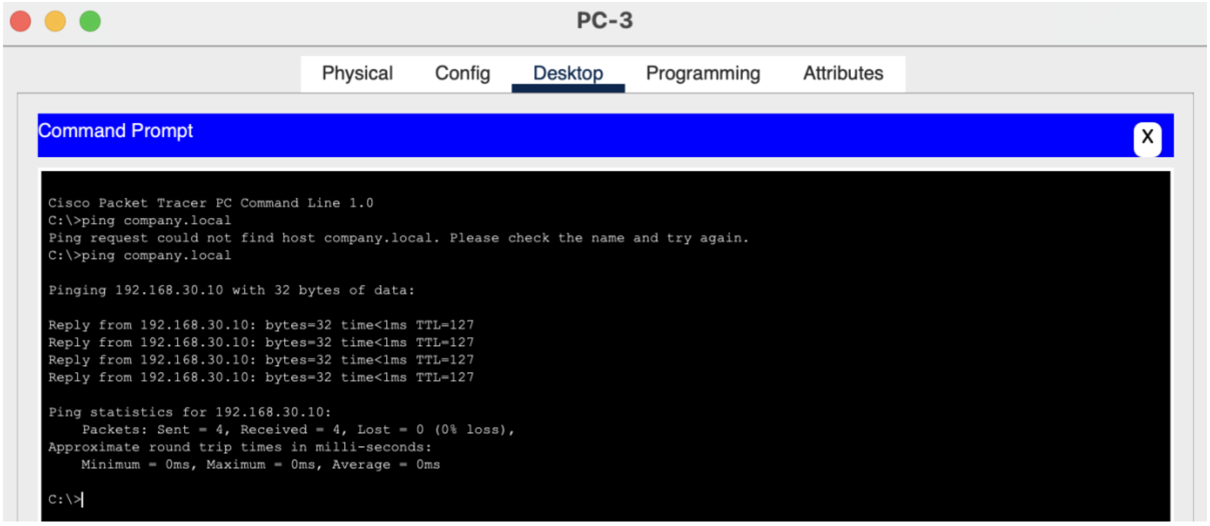
Test 4 – End-to-End Connectivity
PC in VLAN 30:
ping <IP of PC in VLAN 20>



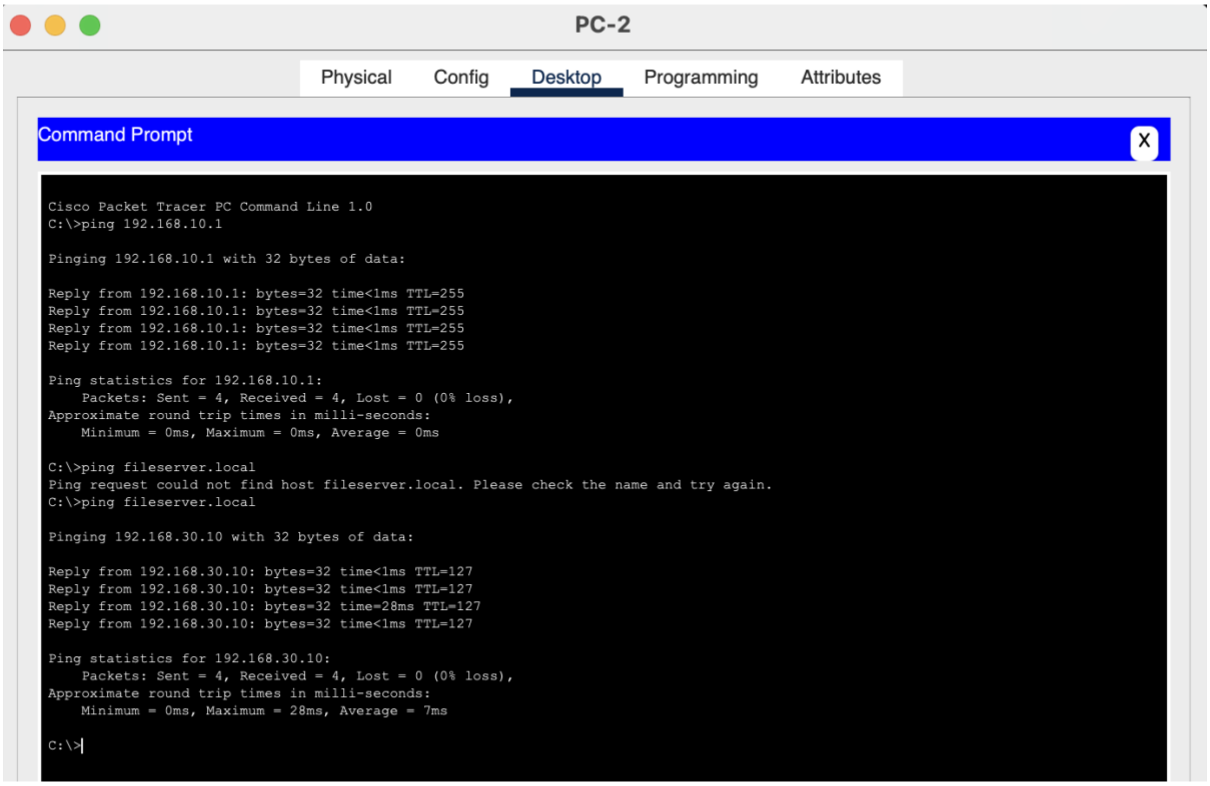
Result: First request timed out but all other packets sent by PC-5 were received by PC-3.

Test 5 – DNS

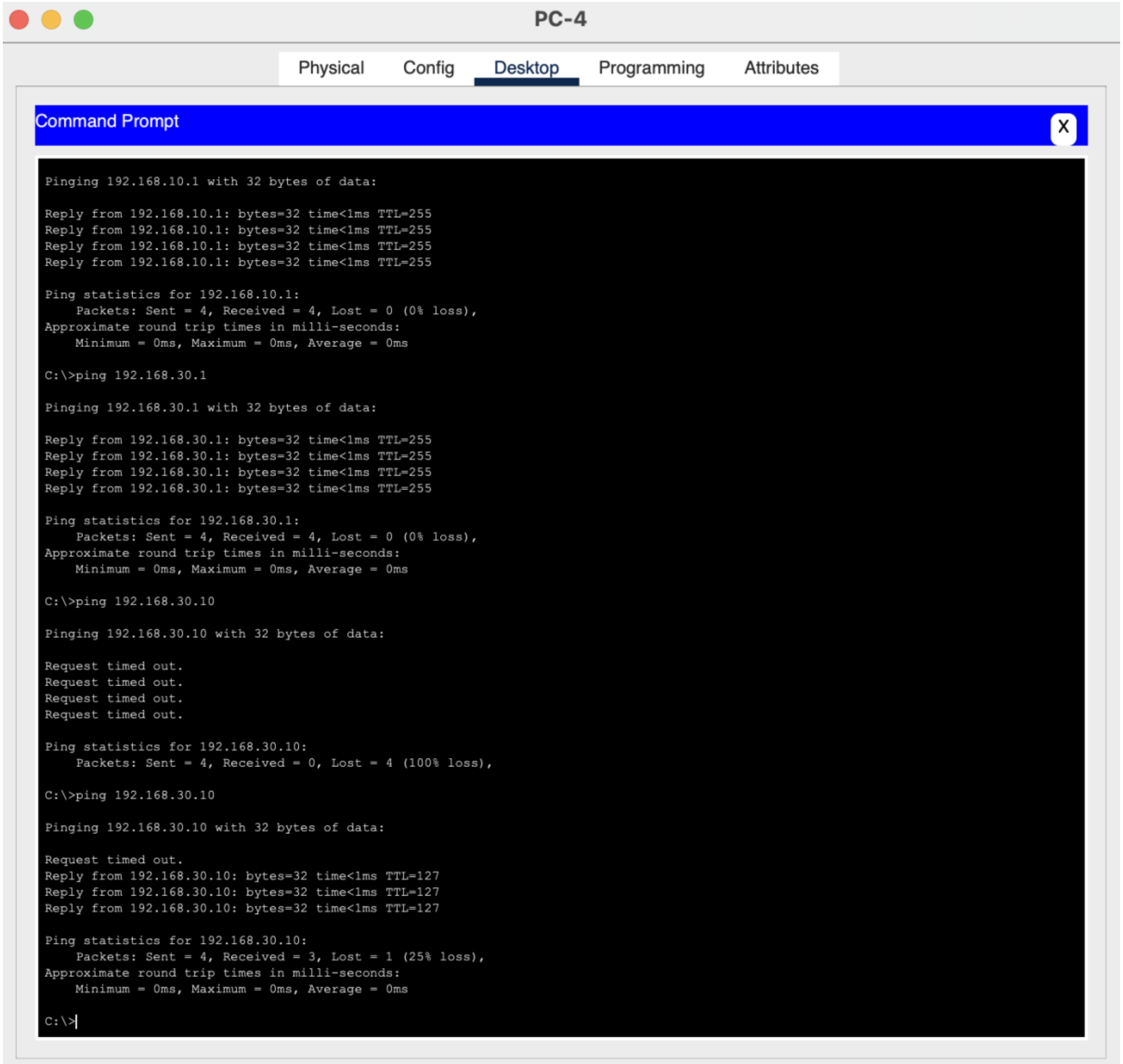
ping company.local



ping fileserver.local



ping 192.168.30.10



Result: The DNS resolves to 192.168.30.10 and multiple PC's are able to ping the DNS Server

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10. Conclusion

The network was successfully configured using VLAN segmentation, trunking, router-on-a-stick inter-VLAN routing, DHCP for dynamic host addressing, and DNS services for hostname resolution. Connectivity tests confirmed successful communication between VLANs and proper DNS functionality.